

LLM-Mediated Human-AI Interaction in Search and Rescue: Impact of Expertise on Attentional Allocation

Human performance, visual attention, and expertise under LLM guidance

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Human-AI Setting in **Critical** Situation

Operators must locate and recover people while conditions change quickly.

- Missions occur in **challenging environments**
- Decisions must be made under **time pressure**
- Need **reliable partner** to help humans ¹



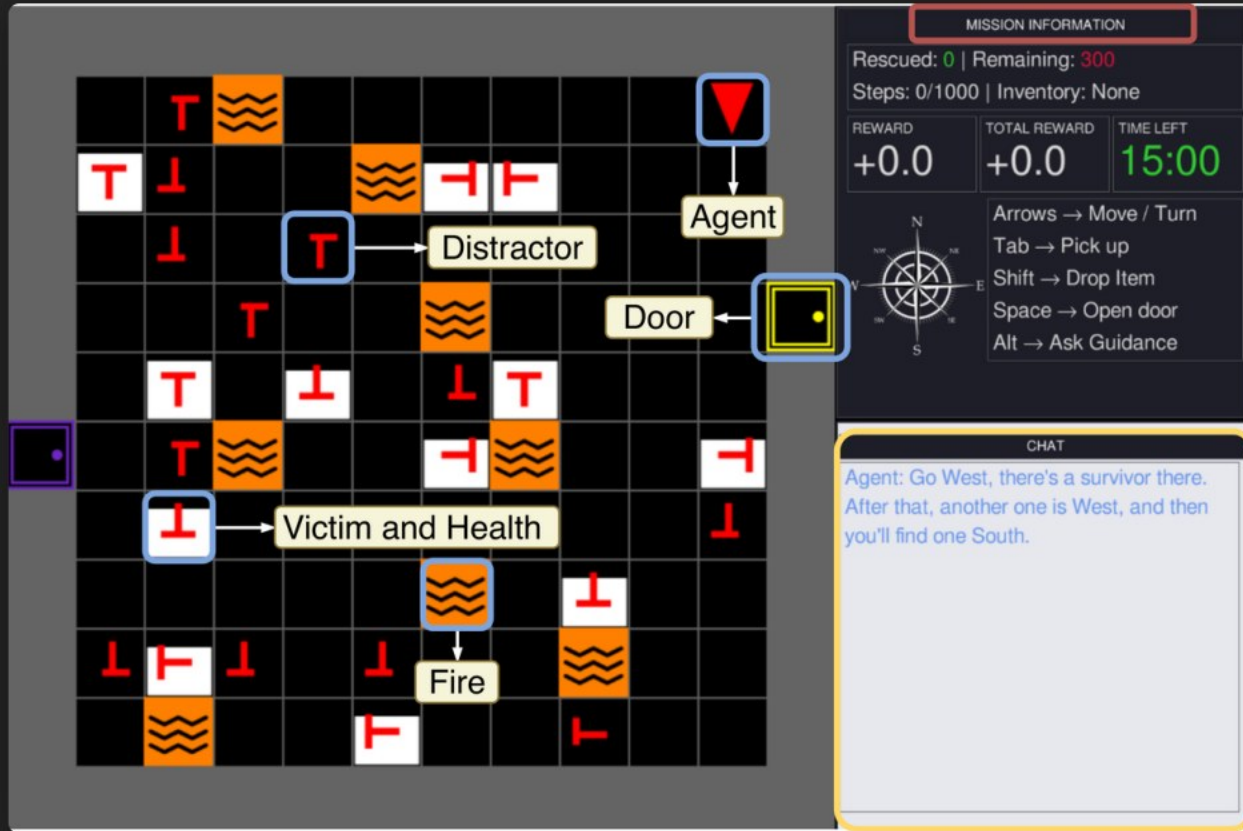
Research Questions and **Contribution**

- **Performance:** Does LLM guidance improve task outcomes?
- **Attention:** How does guidance change visual attention and eye-tracking metrics?
- **Expertise:** Does expertise moderate these effects?
- **Testbed:** A controlled simulation supports repeatable human-AI teaming research.



We study how humans and AI **process information together**, using **eye tracking** to inform **adaptive AI** for the future.

Study Design In Brief



13

PARTICIPANTS

3

TRIALS EACH (RANDOMIZED)

2 + 1

LLM + BASELINE

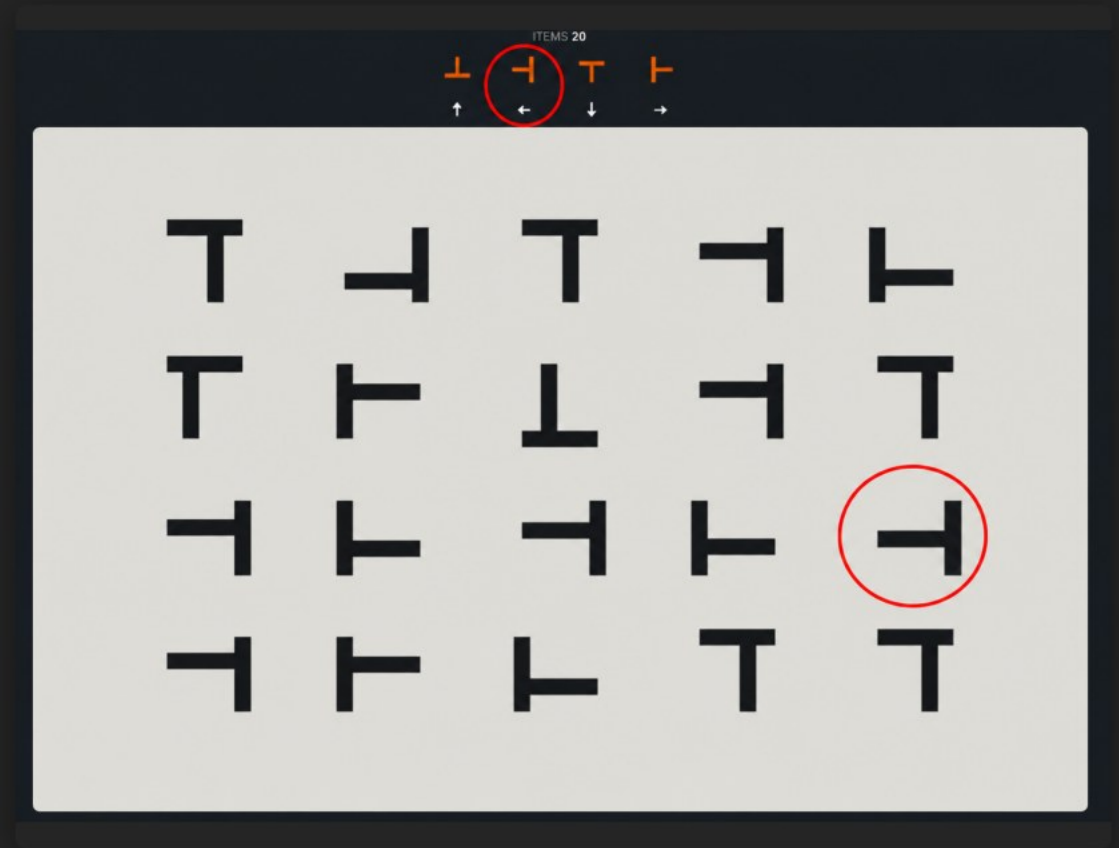
15 min

OR 1,000 MOVES

Task: rescue victims efficiently in a partially observable grid while victim health deteriorates.

ALT requests advice and reveals victim health for **5 seconds**.

Expertise Level



- **Expertise labeling:** K-means clustering ($K = 2$) grouped participants into expert vs. novice²

How The LLM Was Prompted

GUIDANCE STYLE

- You are a search-and-rescue coordinator giving calm, human-like radio guidance to the operator.
- Instructions must be directional and action-focused, always using North, South, East, West, with no coordinates, step counts, or visible reasoning.
- Output either a sparse one-sentence instruction or a detailed 3–4 sentence radio briefing, always concise, natural, and wrapped strictly as <START> ... <END>.

PRIORITY

- Prioritize visible reachable survivors who are still saveable, nearest first.
- From the current state {obs}, choose one immediate objective by prioritizing the nearest reachable visible saveable survivor, then the next saveable survivor if relevant.
- Otherwise choose the best remaining alive victim, then the nearest reachable survivor or key, or finally explore visible areas that may open a new path.

MISSION RULES

- Follow mission and environment rules: rescue all survivors.
- Use matching keys for locked doors, unlocked doors can be opened, carry only one key, avoid lava.
- Never mention unreachable objects. Final instruction must be rule-compliant.

- **Grid-based world:** each cell represents an object such as empty space, wall, victim, lava, door, or key. ³
- **Reachability check:** BFS determines which targets are reachable and their distance from the player. ⁴
- Models: **OpenAI** and **Gemini**.

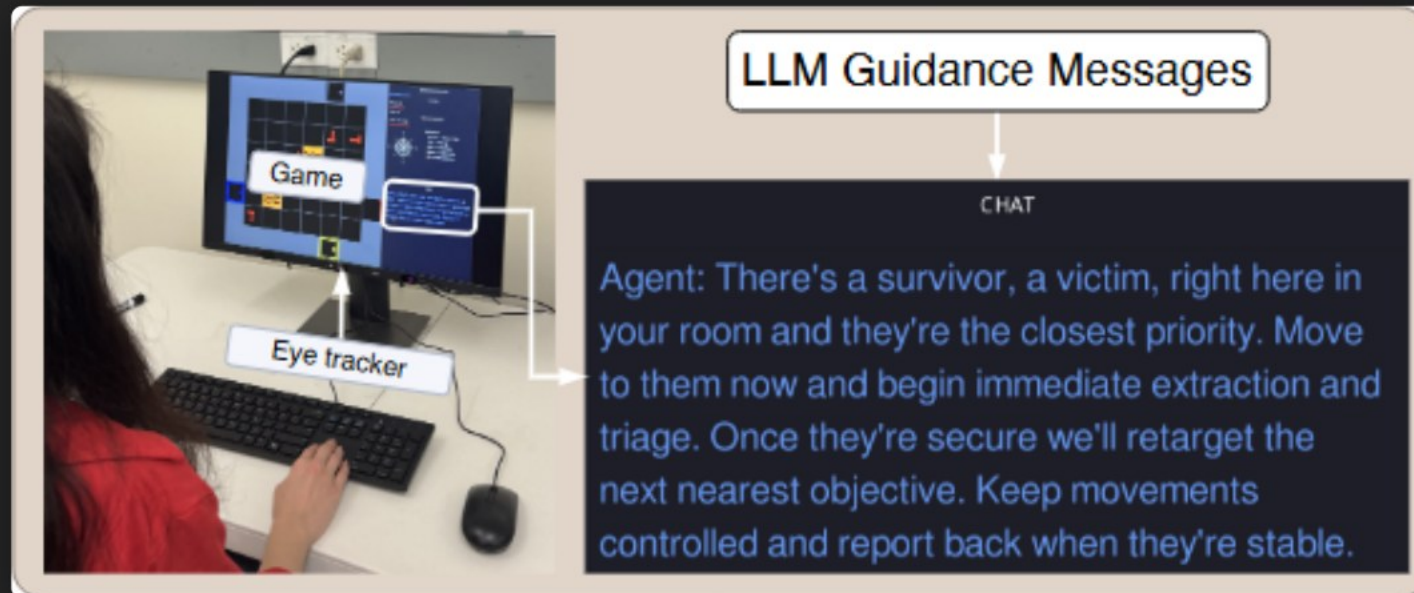
Task and Eye-Tracking Features

FEATURE	WHAT IT SHOWS
Victims per step	Rescue efficiency over time
Total reward	Overall task performance
Saved victims	Mission success
Fixation count	How often the participant focused on an area
Fixation duration	How long the participant looked at one area
Saccade duration	Duration of fast eye movements
Pupil diameter SD	Variation in pupil size, related to workload
Game-area fixation time	Attention to the SAR environment
Chat-panel fixation time	Attention to the LLM guidance

Data Collection and Analysis

- Synchronized data streams via **LSL (Lab Streaming Layer)**
- Gaze and pupil data recorded with a **Tobii eye tracker**

Mixed-effects model: outcome = LLM condition + LLM × expertise + expertise + participant random intercept.



LLM Guidance Improved **Performance**

OUTCOME	EFFECT	β	q
Victims per step	LLM	0.020	< .01
	Expertise	0.016	ns
	LLM $\times$ Expertise	-0.009	ns
Total rewards	LLM	267.143	< .01
	Expertise	-17.714	ns
	LLM $\times$ Expertise	181.857	ns
Saved victims	LLM	0.035	ns
	Expertise	0.221	ns
	LLM $\times$ Expertise	0.435	ns

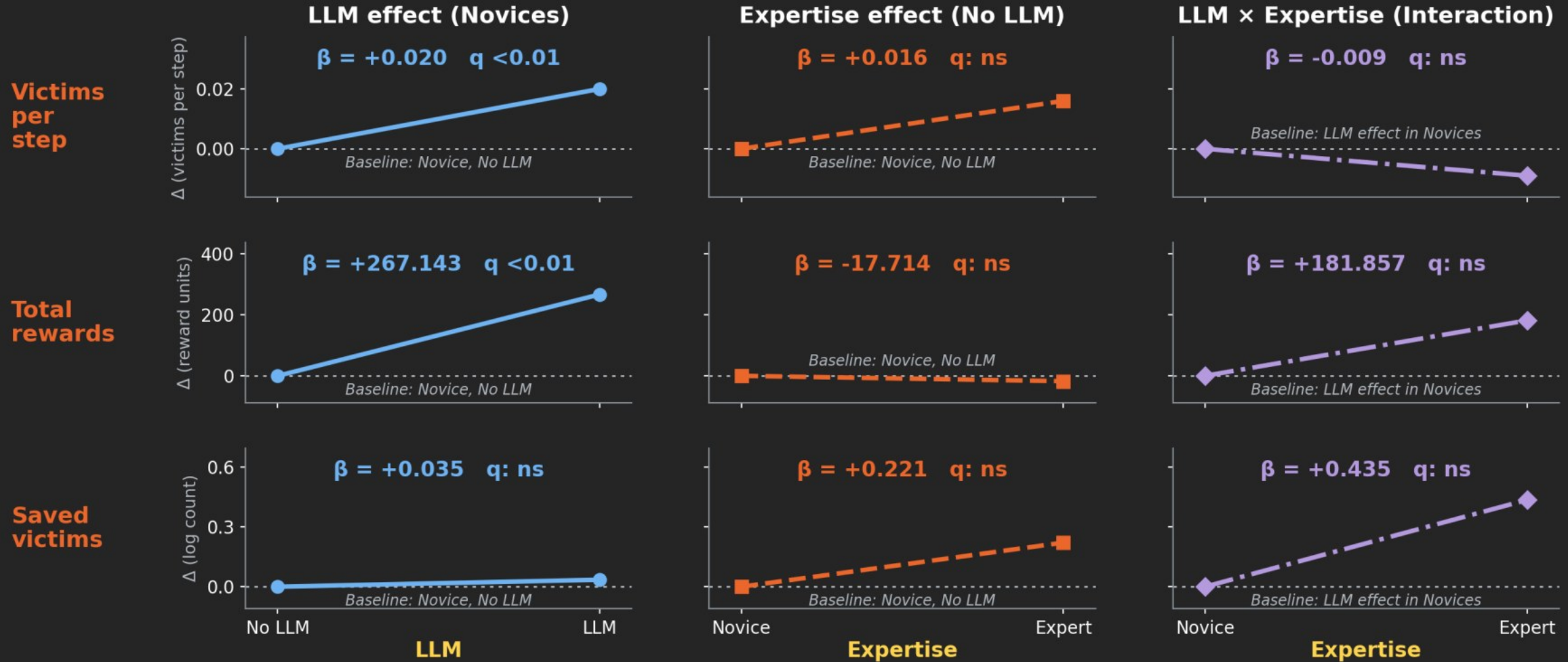
LLM Guidance Changed **Visual Behavior**

OUTCOME	EFFECT	β	q
Fixation duration	LLM	-19.326	< .01
	Expertise	-26.475	ns
	LLM × Expertise	17.059	ns
Pupil-size SD	LLM	0.122	< .0001
	Expertise	-0.042	ns
	LLM × Expertise	-0.051	< .05
Game-area fixation	LLM	-0.184	< .0001
	Expertise	-0.015	ns
	LLM × Expertise	0.114	< .01

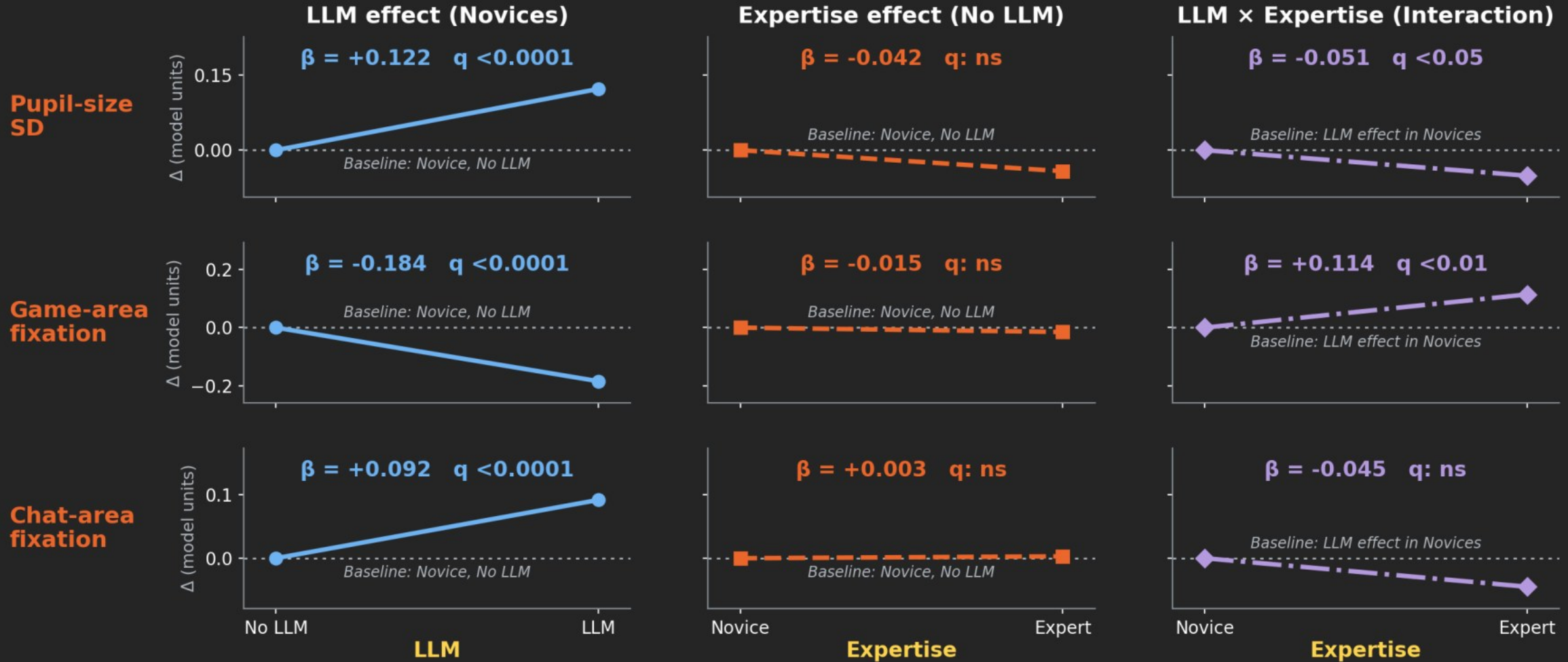
LLM Guidance Changed **Visual Behavior**

OUTCOME	EFFECT	β	q
Chat-area fixation	LLM	0.092	< .0001
	Expertise	0.003	ns
	LLM $\times$ Expertise	-0.045	ns
Fixation count	LLM	0.248	ns
	Expertise	-0.201	ns
	LLM $\times$ Expertise	0.815	< .05
Saccade count	LLM	0.234	ns
	Expertise	-0.482	ns
	LLM $\times$ Expertise	0.961	< .01

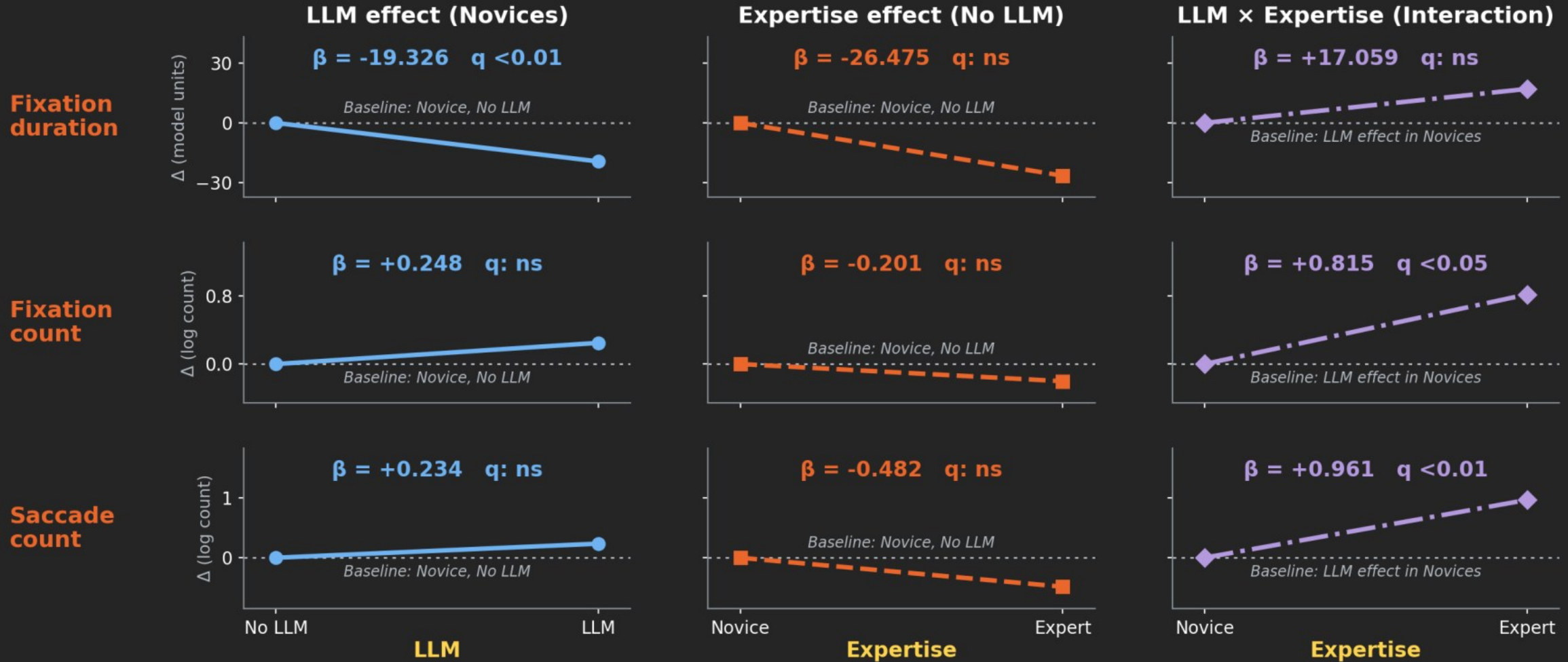
Results: Performance



Results: Pupil and Area Fixation



Results: Fixation and Saccade



Expertise Changes **How Guidance Is Used**

EXPERTS

Maintained more engagement with the game under LLM support and showed stronger changes in visual scanning.

NOVICES

Shifted more attention away from the game, consistent with greater reliance on the guidance panel.

LLM guidance received



Expert

Cross-checks
guidance against the
environment (scans,
verifies)

Novice

Follows guidance
directly

Limitations and Future Work

Limitations

- Sample size
- Simulated grid-world task
- Incorrect/conflicting LLM guidance was not tested
- Additional physiological measures were not included

Future work

TRUST & RELIABILITY

Test how LLM errors affect operator responses and trust.

ADAPTIVE GUIDANCE

Tailor help using performance and physiological signals to reduce mental workload.

SITUATION AWARENESS

Use this data to infer workload, situational awareness, and human-AI interaction patterns.

OTHER DOMAINS

Apply this approach to other human-AI teaming domains beyond search and rescue.

The next step is moving from **static guidance** toward **reliable, adaptive human-AI teaming**.

References

- [1] Handler, Abram, Kai R. Larsen, and Richard Hackathorn. "Large language models present new questions for decision support." *International Journal of Information Management* 79 (2024): 102811.
- [2] J. P. Distefano, H. Manjunatha, S. Chowdhury, K. Dantu, D. Doermann and E. T. Esfahani, "Using Physiological Information to Classify Task Difficulty in Human-Swarm Interaction," *2021 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, Melbourne, Australia, 2021, pp. 1198-1203, doi: [10.1109/SMC52423.2021.9658653](https://doi.org/10.1109/SMC52423.2021.9658653).
- [3] Chevalier-Boisvert, M., Dai, B., Towers, M., Perez-Vicente, R., Willems, L., Lahlou, S., Pal, S., Castro, P. S., & Terry, J. (2023). Minigrid & Miniworld: Modular & Customizable Reinforcement Learning Environments for Goal-Oriented Tasks. *Advances in Neural Information Processing Systems (NeurIPS)*, 36.
- [4] Siklóssy, Laurent, A. Rich, and Vesko Marinov. "Breadth-first search: some surprising results." *Artificial Intelligence* 4.1 (1973): 1-27.



Thank you

Questions?